



AFINMIC

African Inclusive Microscopy



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AFINMIC 2025 International Student Microscopy Symposium

DAY 1, 1 DECEMBER 2025, PARALLEL SESSION A – Light (LM) & Stereomicroscopy (SM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
<u>0</u>	08h 00 - 08h 15	Prof R. Genga	Welcome Address	-
	08h 15 - 08h 45	Prof F. Mücklich	Keynote Talk	
<u>1</u>	08h 45 - 09h 00	Joan Nambafu	Prevalence and economic impact of bovine fasciolosis at the Kampala city abattoir.	Life Science
	09h 00 - 09h 20	Grace wambui Kimani	Evaluation of toxicity, in vivo and invitro efficacy of a medicinal plant extract in malaria disease treatment	Both Material and Life Science
	09h 20 - 09h 40	Chloe Odeh	Neuroprotective Effects Of Caffeine On The Developing Brain And Neurodevelopmental Behaviour Of Kaolin-Induced Hydrocephalic Mice Pups	Life Science
	09h 40 - 10h 00	Joy Dasofunjo	Immunohistochemical evaluation of the hippocampus of alloxan-nicotinamide induced diabetic male wistar rats: The Impacts of Andrographis paniculata	Life Science
	10h 00 - 10h 20	Saffie Bah	Extraction, Optimization and Characterization of Pectin Extracted From Seaweed Using Microwave and Conventional Method	Life Science
	10h 20 - 10h 40	Ayomide Isaac	NEUROPROTECTIVE EFFECT OF TURMERIC ON CALABASH CHALK-INDUCED TOXICITY ON MICE HIPPOCAMPUS	Life Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
<u>2</u>	11h 05 - 12h 05	Ms S. Hendriks	ZEISS	
<u>3</u>	12h 10 - 12h 30	Eisa Houssein	Microscopic Evaluation of Enamel Surface Changes After Teeth Whitening Procedures	Both Material and Life Science
	12hh 30 - 12h 50	Janet Dete	Microscopic Analysis of Pineapple Peel Extracts for the Development of Eco-Friendly Mosquito Repellents	Both Material and Life Science
	12h 50 - 13h 05	Destrial Kamau	Comparative Light Microscopic Characterisation of Selected Mammalian Organs Across Species: Kidney and Spleen (Rabbit), Tongue (Monkey), Testis and Colon (Rat), and Pancreas (Goat)	Life Science
13h 05 - 14h 00			Break	
<u>4</u>	14h 00 - 14h 20	Iheanyi K. Peter-Ajuzie	Detection of West Nile virus in wild captive pteropodid bats and African Giant rats	Life Science
	14h 20 - 14h 40	Yared Wondiye	Title: Monocular Depth-to-3D for Microscopy: Domain-Aware Single-Image	Material Science

			Reconstruction with Multi-Dataset Mixing, Adverse-Translation Consistency, and Semantic Refinement	
	14h 40 - 15h 00	Amal Jamal Balla Ibrahim	Application of Artificial Intelligence to diagnostic and classification ovarian cancer in Sudanese	Life Science
	15h 00 - 15h 20	Hiba Esawy	Cytomorphologic Patterns of Malignant Cell Infiltration into Body Effusions: A Retrospective Microscopy-Based Study	Life Science
	15h 20 - 15h 40	thameen jaradat	Precision Through Vision: The Evolving Role of Microscopy in Neurosurgery and Research.	Both Material and Life Science
	15h 20 - 15h 40	Toheeb Yusuf	Hippocampal Vulnerabilities in Maternal Brains: Effects of Early Onset of Intrauterine Growth Restriction in Female Adult Mice	Life Science
	15h 40 - 16h 00	Session Discussion		
ANNOUNCEMENTS				
END of DAY 1 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 0: <https://wits-za.zoom.us/j/96471623011>

Session 1: <https://wits-za.zoom.us/j/95778942757>

Session 2: <https://wits-za.zoom.us/j/98171351820>

Session 3: <https://wits-za.zoom.us/j/96007543272>

Session 4: <https://wits-za.zoom.us/j/92103786171>

AFINMIC 2025 International Student Microscopy Symposium

DAY 2, 2 DECEMBER 2025, PARALLEL SESSION B - Light (LM) & Stereomicroscopy (SM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
<u>0</u>	08h 00 - 08h 15	Prof R. Genga	Welcome Address	-
	08h 15 - 08h 45	Prof F. Mücklich	Keynote Talk	
<u>5</u>	08h 45 - 09h 00	Vivian Maduabuchi	Neurobehavioral Effects of Naringin On the Hippocampus of Male Swiss Mice Following Exposure To 3,4-Methylenedioxy Methamphetamine (MDMA)	Life Science
	09h 00 - 09h 20	Yusrah Owokoniran	Histological Evaluation of the Neuroprotective Effects of Ginkgo biloba on the Cerebellum of Manganese-Induced Male Wistar Rats Using Light Microscopy	Life Science
	09h 20 - 09h 40	Luimommei Zethan Kpasham	Neurobehavioural, Histological and Immunohistochemical Assessment of Mouse Sciatic Nerve after Sub-Chronic and Chronic Vanadium Exposure	Life Science
	09h 40 - 10h 00	Chioma Njoku	Neuroprotective and Neurobehavioural Effects of Naringin on the Prefrontal Cortex of Adult Swiss Mice exposed to 3, 4-methyzenedioxymethaphetamine (MDMA).	Both Material and Life Science
	10h 00 - 10h 20	Kabeerah Livingstone	Investigating the role of NLRP3 methylation in an in vitro Model: A link between Neuroinflammation and Schizophrenia	Life Science
	10h 20 - 10h 40	Mbawe Zulu	Effect of Hypoxoside on the hippocampus of male Wistar rats exposed to Highly Active Antiretroviral Therapy	Life Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
2	11h 05 - 12h 05	Ms S. Hendriks	ZEISS	
6	12h 10 - 12h 30	Salma Awad	Anopheles arabiensis Patton in Khartoum State: Intraspecific Morphological and Morphometric Variability of Larvae	Both Material and Life Science
	12hh 30 - 12h 50	Alene Admas Birhanu	Isolation and Characterization of Indigenous Wild Yeasts from Local Resources for Bioethanol Production using Molasses	Life Science
	12h 50 - 13h 05	Megan Cleminson	Cellular Architecture of the Nile Crocodile Lung and its Correlation to Adaptational Air-breathing	Life Science
13h 05 - 14h 00			Break	

7	14h 00 - 14h 20	Alsiddig Albashir	Olympus	Both Material and Life Science
	14h 20 - 14h 40	Hoi Ching Cherry Chan	Ultrastructural and Morphological Assessment of ROS-Induced Apoptosis in HeLa Cells Treated with Vitamin D Analogues	Life Science
	14h 40 - 15h 00	Emmanuel Musenge	Cassia abbreviata Oliv. induced pancreatic islet β-cell regeneration in type 2 diabetic male adult Wistar rats	Life Science
	15h 00 - 15h 20	Vincent Wainaina	Hidden Structures, Active Rifts: What Microtextures Reveal About Mt. Kenya and Menengai	Both Material and Life Science
	15h 20 - 15h 40	Abubakar Issack	The Role of Microscopy in Bridging STEM Education Gaps in Africa	Life Science
	15h 40 - 16h 00	Yusrah Owokoniran	Histological Evaluation of the Neuroprotective Effects of Ginkgo biloba on the Cerebellum of Manganese-Induced Male Wistar Rats Using Light Microscopy	Life Science
	16h 00 - 16h 20	Session Discussion		
ANNOUNCEMENTS				
END of DAY 1 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 0: <https://wits-za.zoom.us/j/96471623011>

Session 2: <https://wits-za.zoom.us/j/98171351820>

Session 5: <https://wits-za.zoom.us/j/92637675031>

Session 6: <https://wits-za.zoom.us/j/91763783581>

Session 7: <https://wits-za.zoom.us/j/93041947629>

AFINMIC 2025 International Student Microscopy Symposium

DAY 1, 1 DECEMBER 2025, PARALLEL SESSION C - Light (LM) & Stereomicroscopy (SM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
0	08h 00 - 08h 15	Prof R. Genga	Welcome Address	-
	08h 15 - 08h 45	Prof F. Mücklich	Keynote Talk	
8	08h 45 - 09h 00	Owabo Mkhehle	Light microscopy characterization of gastrointestinal parasitic fauna in mesocarnivore faecal samples: A comparative study across three host species	Life Science
	09h 00 - 09h 20	Mufutau Oladele Oladimeji	A Comparative Evaluation of Traditional Light Microscopy and Digital Microscopy Systems: Assessing Image Quality and Interpretative Efficacy	Life Science
	09h 20 - 09h 40	Ayomide Isaac	Neuroprotective effect of turmeric on calabash chalk-induced toxicity on mice hippocampus	Life Science
	09h 40 - 10h 00	Victory Adekunle	MICROANATOMICAL EFFECTS OF β , ϵ -CAROTENE- 3, 3'-DIOL ON INDOMETHACIN - INDUCED GASTRIC ULCER IN MALE WISTAR RATS	Life Science
	10h 00 - 10h 20	Mohammed Ahmed Ahmed Alhmeem	TBC	Both Material and Life Science
	10h 20 - 10h 40	Eazaldeen Ahammad	Studying Rocks Under A Polarizing Microscope	Material Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
2	11h 05 - 12h 05	Ms S. Hendriks	ZEISS	
9	12h 10 - 12h 30	Carol Gichohi	Antifungal activity of actinomycetes isolates against selected fungal pathogens infecting tomato	Life Science
	12h 30 - 12h 50	Brian Mwangi	Decoding AMR: Visualizing E. coli's Morphological Response to Beta-Lactam Stress	Life Science
	12h 50 - 13h 05	Kingsley Friday	Histological Evaluation of Heavy Metal Toxicity in the Gill, Liver, and Muscle Tissue of Nile Tilapia Fish (Oreochromis niloticus) from a Concrete Pond, Earthen Pond, and a Natural River	Life Science
13h 05 - 14h 00			Break	
10	14h 00 - 14h 20	Susan Omuka	Microscopy as a supporting technique in characterizing foodborne bacteria for antibacterial activity evaluation.	Life Science

	14h 20 - 14h 40	Simon Ndirangu	AI-Enhanced Light Microscopy System	Material Science
	14h 40 - 15h 00	Zinhle Radebe	Cholecalciferol-Induced Cytotoxicity and Morphological Alterations in Cervical Squamous Cell Carcinoma Lines	Life Science
	15h 00 - 15h 20	Zeinab Ahamed Daffaallah	Assessment of cytomorphological changes among sudanese children with sickle cell disease	Life Science
	15h 20 - 15h 40	Christevie Mbuyu	Optimizing Photobiomodulation For Smooth Muscle Differentiation of Adipose-Derived Stem Cells Using Retinoic Acid and TGFβ in a Two-Dimensional Model	Both Material and Life Science
	15h 40 - 16h 00	Arvish Maharaj	Micromorphological and Histochemical Analysis of leaf, stem bark and flower Secretory Structures in Bougainvillea glabra (Nyctaginaceae)	Life Science
	16h 00 - 16h 20	Michelle Rauleka	MICROMORPHOLOGY, ANATOMY, AND HISTO-PHYTOCHEMICAL PROFILING OF SECRETORY STRUCTURES OF Pelargonium tomentosum	Life Science
	16h 20 - 16h 40	Session Discussion		
ANNOUNCEMENTS				
END of DAY 1 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 0: <https://wits-za.zoom.us/j/96471623011>
Session 2: <https://wits-za.zoom.us/j/98171351820>
Session 8: <https://wits-za.zoom.us/j/95225767891>
Session 9: <https://wits-za.zoom.us/j/92611267566>
Session 10: <https://wits-za.zoom.us/j/91499901933>

AFINMIC 2025 International Student Microscopy Symposium

DAY 2, 2 DECEMBER 2025, Fluorescence (FM) / Laser Scanning Confocal Microscopy (LCSM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
11	08h 45 - 09h 00	Ms C. Maepa	Welcome Address	-
12	09h 00 - 09h 20	Oluwabusayo Folarin	Chronic Vanadium Exposure Promotes Aggregation of Alpha-Synuclein, Tau and Amyloid Beta in Mouse Brain	Life Science
	09h 20 - 09h 40	Mubarak Idris	Microscopic characterization of the retinal structure of african agama lizard: a novel model for advancing vision neuroscience research	Life Science
	09h 40 - 10h 00	David Oyeniran	Neuroplastic Responses to Thermal Stress in Caenorhabditis elegans	Life Science
	10h 00 - 10h 20	Oluwapelumi Solagbade	A High-Resolution of MERFISH Atlas Reveals Spatial Segregation of Glial and Neuronal Niches in the Mouse Hippocampus	Life Science
	10h 20 - 10h 40	Ubong Ekpo	Recovery Sleep Reduces Astrocytes and Microglia Expression, and Increases Cell Numbers in the CA3 Region of Adult Male Wistar Rats Exposed to Total Sleep Deprivation	Life Science
	10h 40 - 10h 55	Session Discussion		
	10h 55 - 11h 00	Break		

13	11h 00 - 12h 00	Ms V. Wagner	WIRSAM SCIENTIFIC	
14	12h 05 - 12h 25	Brian Rono	Illuminating Genetic Changes: Microscopy as a Tool to Visualize Mutation Effects	Life Science
	12h 25 - 12h 45	Taidinda Gilbert	Chronic vanadium exposure promotes aggregation of alpha-synuclein, tau and amyloid beta in mouse model	Life Science
	12h 45 - 13h 05	Modisaotsile Lamola	Illuminating the role of HBV Subgenotype A1 in activating invasion and metastasis in hepatocellular carcinoma	Life Science
13h 05 - 14h 00			Break	
15	14h 05 - 14h 25	Catherina T. Shikongo	Comparative Assessment of Sample Preparation Techniques on the Analytical Performance of XRF in Ore Characterization	Material Science
	14h 25 - 14h 45	Abdulrahman Bello	Modeling Schizophrenia Endophenotypes In Drosophila Melanogaster: Effects Of Ketamine On Anxiety, Aggression, Locomotion, Neuroinflammatory And Neurodegenerative Responses	Life Science
	14h 45 - 15h 05	Marne Zeelie	Analysis of autofluorescence and fluorescence in fibrinolytic microclot complexes from patients with inflammatory diseases: using Long COVID as model.	Life Science
	15h 05 - 15h 25	Shannon Swart	Investigating The Presence Of Circulating Cell-Free Mitochondria And The Autoimmune Profile Of Participants With Long Covid And Type 2 Diabetes: A Comparative Study	Life Science

	15h 25 - 15h 45	Michael Cowley	Effect of different unilateral eccentric exercise modes and volumes on satellite cell activation status in the exercised and contralateral unexercised muscle	Life Science
	15h 45 - 16h 05	Kevin Awuonda	Low-Cost Smartphone-Integrated Fluorescence Microscopy For Rapid Detection Of Microplastic Pollution In Kenyan Waterways	Both Material and Life Science
	16h 05 - 16h 25	Ayotomilola Okeya	Microscopy-Based Detection of Oxidative Stress as a Low-Cost Strategy for Addressing Environmental Infertility in Africa	Life Science
	16h 25 - 16h 45	Session Discussion		
ANNOUNCEMENTS				
END of DAY 2 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 11: <https://wits-za.zoom.us/j/98704398304>

Session 12: <https://wits-za.zoom.us/j/96541314424>

Session 13: <https://wits-za.zoom.us/j/95242212779>

Session 14: <https://wits-za.zoom.us/j/99958531606>

Session 15: <https://wits-za.zoom.us/j/93490573186>

AFINMIC 2025 International Student Microscopy Symposium

DAY 3, 3 DECEMBER 2025, PARALLEL SESSION A - Scanning Electron Microscopy (SEM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
16	08h 30 - 08h 45	Ms C. Maepa	Welcome Address	-
17	08h 45 - 09h 00	Senzosenkosi Mkhize	Characterization and Biological Evaluation of Zinc Oxide Nanoparticles Synthesized from Pleurotus ostreatus Mushroom	Life Science
	09h 00 - 09h 20	Amira Mohammed Ali	Solar-Responsive Interpenetrating Biopolymer–Carbon Nanotubes Aerogels for Sustainable Water Capture from Air	Material Science
	09h 20 - 09h 40	Karabo Setsiba Setsiba	The effect of phenothiazine-derivatives on the morphology of the Candida albicans and Candidozyma auris biofilms	Life Science
	09h 40 - 10h 00	Khaleke Veronicah Ramollo	Synthesis and characterisation of Pd@PANI/MIL-101 (Cr) composite for photocatalytic hydrogen production by recovering palladium ions in wastewater	Both Material and Life Science
	10h 00 - 10h 20	Galaletsang Rankapole	The microstructural evolution of Hybrid Polymer–Carbon–Ceramic nanocomposites	Material Science
	10h 20 - 10h 40	Mulugeta Tadesse	Effects of thiourea concentrations on the structural, morphological, and optical properties of lead sulfide thin films prepared by the solution growth method	Material Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
18	11h 05 - 12h 05	Mr M. Kola	SPS	
19	12h 10 - 12h 30	Fatma Ezzahra Dhif	Thickness-Driven Structural Evolution and Interface Modulation in ZnFe ₂ O ₄ -ZnO Nanocomposite Thin Films for Antibacterial Applications	Both Material and Life Science
	12hh 30 - 12h 50	Moyahabo Rebecca Masenya	Evaluating the effects of broad-spectrum bismuth complex hydrogel formulations on biofilm production in burn wounds.	Life Science
	12h 50 - 13h 05	Courtney Hill	Exploring the ex vivo effects of Bitis arietans snake venom on the coagulation, ultrastructure and viscoelastic properties of human blood	Life Science
13h 05 - 14h 00			Break	
20	14h 00 - 14h 20	Fetene Fufa	Preparation and Characterization of Bioactive Glass Nanomaterials for Bone Tissue Engineering	Material Science
	14h 20 - 14h 40	Heinrich Smuts	The ex vivo effects of Naja mosambica venom on blood cell ultrastructure and coagulation in a hypercoagulable model using type 2 diabetes mellitus patients	Life Science

	14h 40 - 15h 00	Otieno Winstone	Remediation Of Fluoride In Contaminated Water Using Unmodified And Modified Fresh Water Oyster Shell Biosorbent	Both Material and Life Science
	15h 00 - 15h 20	Fazeel Ahmad	Integrating Experimental Data with AI-Driven tools for Optimized Kitchen Waste-to-Bio-Diesel	Both Material and Life Science
	15h 20 - 15h 40	Herine Auma	Flood Risk Modelling And Forecasting Around Lake Victoria Kisumu County, Kenya	Material Science
	15h 40 - 16h 00	Fezile Motsoene	Evaluating the Antibacterial Activity of Cerium Oxide Nanoparticles on P. aeruginosa	Both Material and Life Science
	16h 00 - 16h 20	Hajam Ramadan	Microscopic Characterization of Drill Cuttings for Enhanced Drilling Optimization	Material Science
	16h 20 - 16h 40	Session Discussion		
ANNOUNCEMENTS				
END of DAY 3 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 16: <https://wits-za.zoom.us/j/95524470464>

Session 17: <https://wits-za.zoom.us/j/92457225833>

Session 18: <https://wits-za.zoom.us/j/95524470464>

Session 19: <https://wits-za.zoom.us/j/97708576221>

Session 20: <https://wits-za.zoom.us/j/91426773720>

AFINMIC 2025 International Student Microscopy Symposium

DAY 3, 3 DECEMBER 2025, PARALLEL SESSION B - Scanning Electron Microscopy (SEM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
16	08h 30 - 08h 45	Ms C. Maepa	Welcome Address	-
21	08h 45 - 09h 00	Martin Nanje	Microscopy-Guided Artificial Intelligence for Smarter Soil–Sensor Interaction and Precision Irrigation in Semi-Arid Farming Systems	Both Material and Life Science
	09h 00 - 09h 20	Adeniyi Philip Adebule	Microscopy-Guided Insights into Nanoparticle-Mediated Detoxification of Lignocellulosic Hydrolysates for Enhanced Yeast Growth and Bioethanol Production	Both Material and Life Science
	09h 20 - 09h 40	Luchian Pullen	Quantifying dislocation density using Electron Channeling Contrast Imaging (ECCI).	Material Science
	09h 40 - 10h 00	Hamza El Fadili	Engineering performance of binary and ternary alkali-activated materials incorporating phosphate sludge and sugar lime: A by-product valorisation perspective	Material Science
	10h 00 - 10h 20	Raphael Evans	Microscopic Insights into Soil Microstructure: Scanning Electron Microscopy (SEM) as a Tool for Assessing Soil Stability and Environmental Resilience in Degraded Landscapes	Material Science
	10h 20 - 10h 40	Kayode Apo	Morphological study of the leaf of Ocimum americanum (L.) to identify their trichomes and biochemical constituents	Life Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
18	11h 05 - 12h 05	Mr M. Kola	SPS	
22	12h 10 - 12h 30	Sinesipho Phalaso	Morphological and Cytocompatibility Analysis of Polycaprolactone and Gelatin Nanofiber Mats.	Life Science
	12hh 30 - 12h 50	Velaphi Moyo	Microtextural and Elemental Characterization of Low-Grade Sulphide and Oxide Gold Ores Using SEM-EDS	Material Science
	12h 50 - 13h 05	Abdelhadi Zouhair	Poly(ϵ -caprolactone) Grafted from Hydrohydroxymethylated Vegetable Oils as Biodegradable Coatings for Water-Soluble Fertilizers	Material Science
13h 05 - 14h 00			Break	
23	14h 00 - 14h 20	Tshifhiwa Tshikosi	MICRO-MORPHOLOGICAL AND HISTOCHEMICAL CHARACTERIZATION OF THE SECRETORY STRUCTURES IN LEAVES AND STEMS OF <i>Lantana rugosa</i> L. (VERBENACEAE)	Life Science
	14h 20 - 14h 40	Brian Kariithi	3D Neuro-Mimetic Platforms for Therapeutic Intervention of Glioblastoma Multiforme	Both Material and Life Science

	14h 40 - 15h 00	Mohammed DOUDOUH	Study and Valorization of Metallurgical Slag and Fly Ash for Thermal Energy Storage	Material Science
	15h 00 - 15h 20	Abdelhadi Zouhair	Polymers Based on Hydrohydroxymethylated Vegetable Oils as Biodegradable Coatings for Water-Soluble Fertilizers	Material Science
	15h 20 - 15h 40	Siphosenkosi Makhoba	Lung Cancer Detection Using Vision Transformers	Life Science
	15h 40 - 16h 00	Ashish Misra	Macromorphological, Micromorphological and Silver Nanoparticle synthesis of leaves and stems of Ipomea pes-caprae	Life Science
	16h 00 - 16h 20	Phebo Tebogo Sebobi	Investigating Corrosion Mechanisms and The Electrochemical Performance of Ferrosilicon for Dense Medium Separation Operations in Botswana	Material Science
	16h 20 - 16h 40	Session Discussion		
ANNOUNCEMENTS				
END of DAY 3 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 16: <https://wits-za.zoom.us/j/95524470464>

Session 18: <https://wits-za.zoom.us/j/95524470464>

Session 21: <https://wits-za.zoom.us/j/98103277696>

Session 22: <https://wits-za.zoom.us/j/92779620533>

Session 23: <https://wits-za.zoom.us/j/92371825551>

AFINMIC 2025 International Student Microscopy Symposium

DAY 3, 3 DECEMBER 2025, PARALLEL SESSION C - Scanning Electron Microscopy (SEM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
16	08h 30 - 08h 45	Ms C. Maepa	Welcome Address	-
24	08h 45 - 09h 00	Sohani Amratlal	Micromorphological and histochemical analysis of Coleus amboinicus Lour. (Lamiaceae).	Life Science
	09h 00 - 09h 20	Abdelali Kamouni	Encapsulation of Beauveria bassiana conidia in carboxymethylcellulose beads: Advancing UV protection and thermal resilience in bioinsecticide applications	Both Material and Life Science
	09h 20 - 09h 40	Nompumelelo Nkosi	The Effect Of Casting Methods On The Development Of Low-Density Stainless Steels For Biomedical Implants	Material Science
	09h 40 - 10h 00	Takudzwa Sande	Microscopy Driven Evaluation of Cold Sprayed and Powder Coated Aluminium and Mild Steel Surfaces	Material Science
	10h 00 - 10h 20	Bethaba Shazi	Investigating the Mycobacterium tuberculosis-autophagy niche with volume electron microscopy	Life Science
	10h 20 - 10h 40	Lebogang Mphahlele	Valorization of Eucalyptus grandis residues into green-synthesized mesoporous biochar–silver nanocomposites for azo dye remediation	Material Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
18	11h 05 - 12h 05	Mr M. Kola	SPS	
25	12h 10 - 12h 30	Ndeye Fatou Diop	Hydrothermal synthesis of enhanced zeolitic imidazole frameworks integrated with graphene carbon nitride	Material Science
	12h 30 - 12h 50	Dimpo Sinesipo Sipuka	Application of TiO_2 -1 Magnéli nanofibers towards photoelectrocatalytic degradation of ciprofloxacin in water/wastewater	Material Science
	12h 50 - 13h 05	Helena Zelle	Title: Scanning Electron Microscopy-Based Characterisation of Ex Vivo Blood Component Alterations Induced by Paclitaxel and a Carbazole Derivative	Life Science
13h 05 - 14h 00			Break	
26	14h 00 - 14h 20	Husam Mohammed Ahmed Dfalla	Gas Lift Systems	Both Material and Life Science
	14h 20 - 14h 40	Simon Ndirangu	AI-Enhanced Light Microscopy System	Material Science

	14h 40 - 15h 00	Zinhle Radebe	Cholecalciferol-Induced Cytotoxicity and Morphological Alterations in Cervical Squamous Cell Carcinoma Lines	Life Science
	15h 00 - 15h 20	Zeinab Ahamed Daffaallah	Assessment of cytomorphological changes among sudanese children with sickle cell disease	Life Science
	15h 20 - 15h 40	Christevie Mbuyu	Optimizing Photobiomodulation For Smooth Muscle Differentiation of Adipose-Derived Stem Cells Using Retinoic Acid and TGFβ in a Two-Dimensional Model	Both Material and Life Science
	15h 40 - 16h 00	Tshepo Mohlala	Microscopy-Driven Structural and Functional Analysis of BTO/SIS 0D/3D architectural Hybrids for Pharmaceutical Degradation via piezo-photocatalysis	Material Science
	16h 00 - 16h 20	Absalom Kambonde	Effect of MIG-Weldingf Parameters on Corrosion Behavior of AISI-316 Stainless Steel in Marine Environment	Material Science
	16h 20 - 16h 40	Abraham Okiro	High-Resolution Satellite and Drone-Based Imaging for Monitoring Agricultural Productivity and Climate Stress in Rural Kenya	Material Science
	16h 20 - 17h 00	Session Discussion		
ANNOUNCEMENTS				
END of DAY 3 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 16: <https://wits-za.zoom.us/j/95524470464>

Session 18: <https://wits-za.zoom.us/j/95524470464>

Session 24: <https://wits-za.zoom.us/j/98651301280>

Session 25: <https://wits-za.zoom.us/j/92609030988>

Session 26: <https://wits-za.zoom.us/j/95390973878>

AFINMIC 2025 International Student Microscopy Symposium

DAY 4, 4 DECEMBER 2025, PARALLEL SESSION A – Transmission Electron Microscopy (TEM)

Session	Time Slot (SAST)	Speaker	Title	Discipline
27	08h 30 - 08h 45	Ms C. Maepa	Welcome Address	-
28	08h 45 - 09h 00	Mopholosi Raymond Monnaatsheko	Improving the efficiency, absorption and stability of an organic photovoltaic solar cell (OPV) using TEM guided morphological optimization	Material Science
	09h 00 - 09h 20	Awadelkarim Mohamed Ismail Ibrahim	Physical Properties of Nano- Scaled silver in Boron Nitride Matrix	Material Science
	09h 20 - 09h 40	Nguutor Simon Akiiga	Synthesis and Characterization of Graphene-based Heterostructures for Electrochemical Sensing Applications	Material Science
	09h 40 - 10h 00	Khadija Mohammed Musa	Activated Carbon Production From Pistachio Shells For Use in Supercapacitor Applications	Material Science
	10h 00 - 10h 20	Mohlokamedi Bopape	An in vivo and in vitro examination on the effect of Streptozotocin on rat liver tissue and human hepatocytes	Life Science
	10h 20 - 10h 40	Ngutor Akiiga	Fabrication of graphene-based sensors	Material Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
29	11h 05 - 12h 05	Dr B. Barkay		
30	12h 10 - 12h 30	Ntando Lele	Engineering a novel S-scheme MoS ₂ /WO ₃ @Ti ₃ C ₂ Tx MXene ternary nanocomposite with excellent physicoelectrochemical and photocatalytic properties.	Material Science
	12h 30 - 12h 50	Firdausi Aliyu	Microscopy Meets Genomics: A Hybrid model for Environmental Tracking of Enteroviruses in Kano State, Northwestern Nigeria.	Life Science
	12h 50 - 13h 05	Busisiwe Nomnganga	Purity Assessment of Gardnerella vaginalis Cultures and Ultrastructural Characterization of Their Extracellular Vesicles Using Microscopy.	Life Science
13h 05 - 13h 20			Break	
31	14h 00 - 14h 20	Tegani Ahmed	Effects and consequences of soil pollution through factories and chemical fertilizers	Both Material and Life Science
	14h 20 - 14h 40	John-Andrew Odhiambo	Quantitative Characterization of AL Amyloid Fibril Morphology Using Open Cryo-Electron Microscopy Data (EMPIAR-10245)	Life Science

	14h 40 - 15h 00	Tanaka Kimberly Magunda	Functionalised Zeolitic Immidiazolate framework incooperated with Gold nanoclusters for the selective desulphurisation of fuel oil.	Material Science
	15h 00 - 15h 20	Pierre Palm	Obtaining uniformity: making more uniform biogenic gold nanoparticles from South African sweet potato extract	Both Material and Life Science
	15h 20 - 15h 40	Mohamed TCHIKA	Multifunctional composite magnetic nanoparticles based MnFe2O4 for wastewater treatment	Material Science
	15h 40 - 16h 00	Sachin Bhoora	Polyploid Giant Cells Are Resistant To The Anti-Cancer Action Of Cholecalciferol And Calcidiol In Cervical Cancer Cell Lines	Life Science
	16h 00 - 16h 20	Anelisiwe Gqumani	Investigating the Influence of Terbium Additions on the Microstructure of Quenched Fe-Ga Alloys	Material Science
	16h 20 - 16h 40	Hamza ELJABBAR	Advanced microscopic study of carboxymethyl cellulose (CMC) anchored metal nanoparticles	Material Science
	16h 40 - 17h 00	Noumane Elouakassi	Argan-Waste-Derived Carbon Dots for UV-Blocking Nanocomposites and Solar Cell Performance Enhancement	Material Science
	17h 40 - 17h 20	Session Discussion		
ANNOUNCEMENTS				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 27: <https://wits-za.zoom.us/j/94199976736>

Session 28: <https://wits-za.zoom.us/j/99394981735>

Session 29: <https://wits-za.zoom.us/j/94520204729>

Session 30: <https://wits-za.zoom.us/j/95403319584>

Session 31: <https://wits-za.zoom.us/j/95935299962>

AFINMIC 2025 International Student Microscopy Symposium

DAY 4, 4 DECEMBER 2025, PARALLEL SESSION B – Spectroscopy & Other Techniques

Session	Time Slot (SAST)	Speaker	Title	Discipline
27	08h 30 - 08h 45	Ms C. Maepa	Welcome Address	-
32	08h 45 - 09h 00	Cedrick Nsengiyumva	Biosignatures from the 2.5 Ga Old Stromatolites of the Ramonnedi Formation (Botswana) and their Astrobiological Implications	Life Science
	09h 00 - 09h 20	Habiba Abdelmoniem Hossni	Green synthesis of magnetic Fe ₃ O ₄ /chitosan nanoparticles for efficient adsorption	Material Science
	09h 20 - 09h 40	Adebola Adejumo	TBC	Both Material and Life Science
	09h 40 - 10h 00	Brian Kipruto	Spectroscopic Characterization of Flexible Graded-Z Laminates Using X-Ray Fluorescence (XRF) for Computational Model Validation	Material Science
	10h 00 - 10h 20	Olukemi Daramola	Spectroscopic and Microscopic Characterisation of Photobiomodulation Induced Differentiation of Immortalised Adipose-Derived Stem Cells into Functional Insulin-Producing Beta Cells	Life Science
	10h 20 - 10h 40	Gashaw Erku	Exploring the therapeutic potential of <i>Kirkia tenuifolia</i> Engl. stem bark extract and its bioactive compounds through experimental and computational approaches	Life Science
	10h 40 - 10h 55	Session Discussion		

10h 55 - 11h 05			Break	
33	11h 05 - 12h 05	Mr D. Perrett	Angstrom	
34	12h 10 - 12h 30	Lebogang Ramarope	Green synthesis of carbon dots with antiviral properties against SARS CoV-2 using water and methanol extracts of Erythrina caffra	Both Material and Life Science
	12hh 30 - 12h 50	Petrus Kankonda	Neural Network Based Microstructure Image Analysis Model For Predicting Thermal History In Annealed Mild Steels	Material Science
	12h 50 - 13h 05	Adlight Alivitsa Magui	Deep Learning–Driven Fusion of LIBS and Raman Spectrometry for Rare Earth Element Detection in Radiothermic Carbonatites	Material Science
13h 05 - 14h 00			Break	
35	14h 00 - 14h 20	Lindiwe Maphalla	Evaluating Antibody-Coated Gold Nanoparticles for Rapid LFA-Based Pathogen Detection	Life Science
	14h 20 - 14h 40	Rania Emara	Optical, electrical, thermal and kinetic studies for some pyrimidine ligands and their complexes	Material Science

	14h 40 - 15h 00	Olaide Olarewaju	Comparative phytochemical profiling, assessment of mining-related pollutants in Newbouldia laevis Leaves: A combined GC-MS / FTIR approach	Life Science
	15h 00 - 15h 20	Loubna EL Fedyl	Benzophenone - Copper dual photocatalysis supported on classical and mesoporous silica for the anilines α -amino C–H bonds alkylation	Material Science
	15h 20 - 15h 40	Thato Masilompane	Development of g-C ₃ N ₄ –BiVO ₄ @NiFe-LDH composites for photoelectrocatalytic (PEC) hydrogen generation	Material Science
	15h 40 - 16h 00	Aphiwe Zulu	Chemical Composition of Ekebergia Capensis and its Role in Silver Nanoparticle Synthesis for Severe Syndrome Coronavirus-2 Potential Treatment	Material Science
	16h 00 - 16h 20	Session Discussion		
ANNOUNCEMENTS				
END of DAY 4 PROGRAMME				

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 27: <https://wits-za.zoom.us/j/94199976736>

Session 32: <https://wits-za.zoom.us/j/98992085700>

Session 33: <https://wits-za.zoom.us/j/99944628753>

Session 34: <https://wits-za.zoom.us/j/91857963696>

Session 35: <https://wits-za.zoom.us/j/93516568150>

AFINMIC 2025 International Student Microscopy Symposium

DAY 4, 4 DECEMBER 2025, Awards Ceremony

Session	Time Slot (SAST)	Speaker	Title
36	17h 30 - 18h 00	AFINMIC TEAM	Awards Ceremony

Links to Sessions (Note that the Session Number is hyperlinked to the ones below)

Session 36: <https://wits-za.zoom.us/j/93559581383>